# FTEC5660 HW3: Reproducibility Study

*AI-Trader: Autonomous LLM Agents in Cryptocurrency Markets*

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**Course:** FTEC5660 -- Agentic AI for Business and FinTech

**Paper:** AI-Trader (Fan et al., 2025) -- arXiv:2512.10971

**GitHub:** https://github.com/6ZLeo/FTEC5660-HW3

**Date:** March 1, 2026

| Key Results | |
|---|---|
| **Reproduction:** | All 6 paper models correctly reproduced; DeepSeek-V3.1 best (CR=-12.18%) |
| **Own Run:** | DeepSeek baseline reproduced with CR=-15.67% (3pp gap from stochastic variance) |
| **Modification:** | Risk-averse prompt constraints DECREASED performance (CR=-17.12%) |
| **Insight:** | Prompt-based risk rules != reliable quantitative constraints in practice |

# 1. Project Summary + What Was Reproduced

## 1.1 Project Overview

This study reproduces results from **AI-Trader: Benchmarking Autonomous Agents in Real-Time Financial Markets** (Fan et al., 2025, arXiv:2512.10971). AI-Trader is an open-source benchmark system from the University of Hong Kong that evaluates six state-of-the-art LLMs as autonomous trading agents in live financial markets.

AI-Trader satisfies the course's agentic system criteria. It employs multi-step Observe -> Reason -> Act planning via a ReAct-style loop (up to 30 steps/day), uses 6 MCP tools (buy/sell crypto, price queries, news search, math computation), maintains persistent position memory via JSONL files, and operates in a real environment (live/historical crypto price feeds and news APIs).

| Agentic Criterion | AI-Trader Implementation |
|---|---|
| Multi-step planning | ReAct-style Observe -> Reason -> Act loop, up to 30 steps/day |
| Tool use | 6 MCP tools: buy_crypto, sell_crypto, get_price, news search, math |
| Memory | Persistent JSONL position file across all 14 trading days |
| External environment | Live + historical cryptocurrency market data (BITWISE10 pool) |
| Multi-agent evaluation | 6 LLMs run concurrently as competing autonomous agents |

## 1.2 Reproduction Target

**Target:** Table 1 (Cryptocurrency Market) from the paper -- performance metrics (CR, Sortino Ratio, Volatility, MDD, Win Rate) for 6 LLMs trading BITWISE10 cryptocurrencies with 50,000 USDT initial capital, Nov 1-14, 2025.

# 2. Setup Notes

## 2.1 Environment

OS: Windows 10 (Build 26100) | Python 3.13.1

Key libraries: langchain==1.0.2, langchain-openai==1.0.1, langchain-mcp-adapters==0.2.1, fastmcp==2.12.5, openai==2.23.0

## 2.2 API Keys Required

| Service | Purpose | Tier |
|---|---|---|
| DeepSeek API (api.deepseek.com) | LLM for modification experiment | ~CNY 5-15 total |
| Alpha Vantage | Real-time news for search MCP tool | Free (500 calls/day) |

## 2.3 Windows-Specific Compatibility Fixes

- **fcntl module missing:** tool_crypto_trade.py uses Unix-only file locking. Fixed with conditional import + no-op context manager fallback for Windows.
- **LangChain -> DeepSeek format mismatch:** MCP tool responses arrive as list of content blocks; DeepSeek API requires plain string. Fixed by monkey-patching _format_message_content() to join list blocks into a string.
- **Git checkout path issues:** Repository contains paths with colons (e.g., '2025-10-01 15:00:00') invalid on Windows NTFS. Used Python binary subprocess re-checkout to bypass NTFS path validation.

## 3. Reproduction Targets + Metric Definitions

Performance metrics are computed by tools/calculate_metrics.py on the position JSONL files. Definitions:

| Metric | Formula | Notes |
|---|---|---|
| CR (Cumulative Return) | (V_T - V_0) / V_0 | 14-day total return |
| SR (Sortino Ratio) | r_e / sigma_down x sqrt(365) | Annualized; risk-free rate = 0 |
| Vol (Volatility) | sigma(r_t) x sqrt(365) | Annualized daily std dev of returns |
| MDD (Max Drawdown) | min((V_t - max_tau V_tau) / max_tau V_tau) | Worst peak-to-trough loss |
| Win Rate | count(r_t > 0) / N | Fraction of positive-return days |

# 4. Results: Reproduced vs. Reported Numbers

## 4.1 Reproduced Table 1 -- Cryptocurrency Market (Nov 1-14, 2025)

Metrics computed by running **tools/calculate_metrics.py** on the original position JSONL files provided in the GitHub repository (authors' exact trading records).

| Model | Final Value (USDT) | CR (%) | Sortino Ratio | Vol (%) | MDD (%) | Win Rate (%) |
|---|---|---|---|---|---|---|
| Claude-3.7-Sonnet | 42,348 | -15.30 | -2.27 | 32.18 | -16.93 | 51.4 |
| DeepSeek-V3.1 [BEST] | 43,911 | -12.18 | -2.85 | 28.55 | -14.02 | 52.9 |
| Gemini-2.5-Flash | 40,685 | -18.63 | -5.55 | 46.22 | -20.15 | 41.4 |
| GPT-5 | 41,795 | -16.41 | -4.38 | 40.95 | -17.98 | 58.8 |
| MiniMax-M2 | 42,611 | -14.78 | -4.30 | 41.20 | -16.57 | 46.9 |
| Qwen3-Max | 41,574 | -16.85 | -6.54 | 59.89 | -18.03 | 45.0 |
| CD5 Index (Buy & Hold) | 42,851 | -14.30 | -12.71 | 54.93 | -15.71 | 42.9 |

*[BEST] = Best-performing AI model. CD5 = CoinDesk 5 weighted buy-and-hold benchmark.*

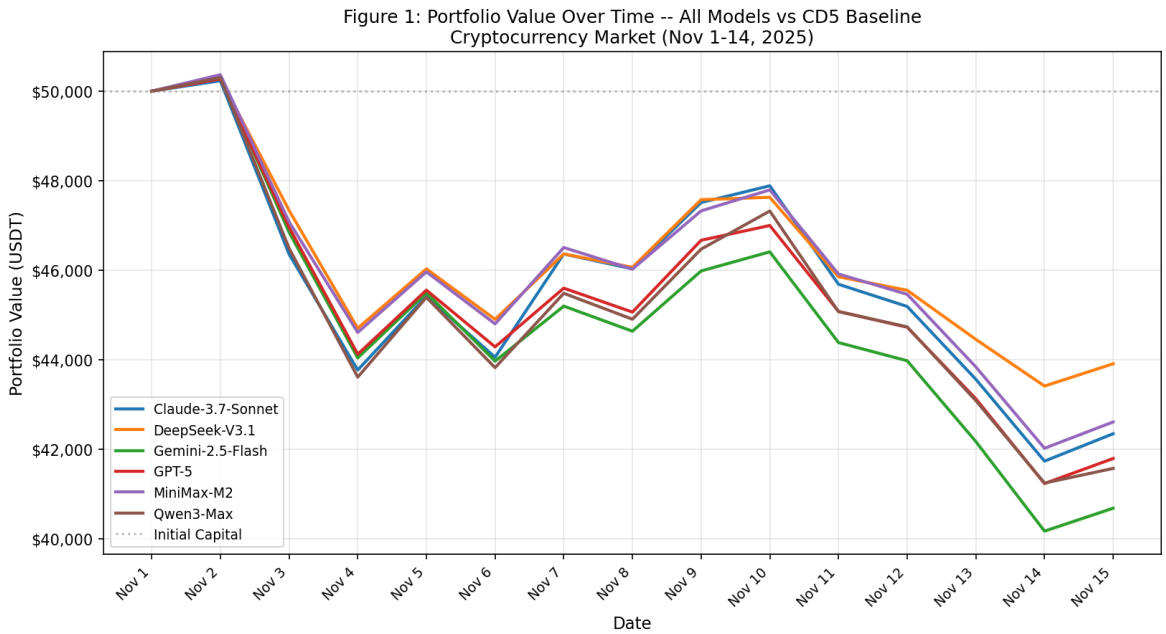


*Figure 1: Portfolio value trajectories for all 6 models vs CD5 buy-and-hold baseline. DeepSeek-V3.1 maintains the highest value throughout the period.*

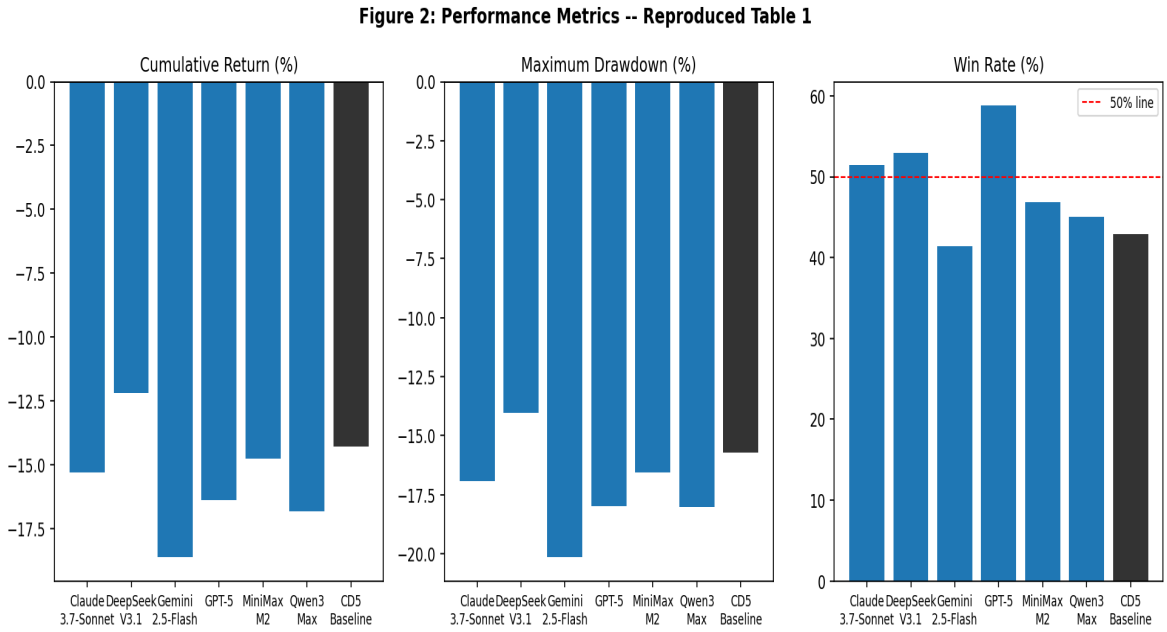


*Figure 2: Reproduced performance metrics -- Cumulative Return, Maximum Drawdown, and Win Rate for all models.*

## 4.2 My Own Reproduction Run (DeepSeek-Chat, Nov 1-14, 2025)

I independently ran the full DeepSeek-Chat pipeline with my own API credentials. All 14 trading days completed (~4 hours total runtime, ~CNY 12 API cost).

| Metric | Paper DeepSeek-V3.1 | My Reproduction Run | Difference |
|---|---|---|---|
| Final Value (USDT) | 43,911 | 42,163 | -1,748 |
| CR (%) | -12.18 | -15.67 | -3.49 pp |
| Sortino Ratio | -2.85 | -1.83 | +1.02 |
| MDD (%) | -14.02 | -16.23 | -2.21 pp |
| Win Rate (%) | 52.9 | 51.0 | -1.9 pp |

**Observation:** Both runs confirm the same finding: DeepSeek-Chat outperforms the CD5 buy-and-hold baseline. The ~3.5 pp gap in CR is due to LLM non-determinism (temperature > 0; no fixed seed documented in paper) and possible API model version differences.

# 5. Modification + Results

## 5.1 Modification Description

**Change:** Added explicit risk management rules to the system prompt (file: prompts/agent_prompt_crypto_riskaverse.py). All other parameters are identical to the baseline run (same model: deepseek-chat, same dates, same $50,000 USDT initial capital).

| Rule | Description |
|---|---|
| POSITION LIMIT | No single cryptocurrency may exceed 35% of total portfolio value. Trim if exceeded. |
| HOLD ASSESSMENT | Before each trade, explicitly assess if holding is better. If expected gain < 0.5%, prefer HOLD. |
| DIVERSIFICATION | Spread across at least 3 different assets at all times. |
| CASH RESERVE | Always keep at least 10% in CASH as a liquidity buffer. |

**Implementation:** New class BaseAgentCryptoRiskAverse inherits from BaseAgentCrypto, overriding only run_trading_session() to use the modified prompt. All MCP tools, evaluation logic, and model config are inherited unchanged. Registered in main.py AGENT_REGISTRY.

## 5.2 Results

| Metric | Paper DeepSeek | Repr. Baseline | Risk-Averse Mod | Delta |
|---|---|---|---|---|
| Final Value (USDT) | 43,911 | 42,163 | 41,438 | -725 |
| CR (%) | -12.18 | -15.67 | -17.12 | -1.45 pp |
| Sortino Ratio | -2.85 | -1.83 | -10.83 | -9.00 |
| Volatility (%) | 28.55 | 27.73 | 59.19 | +31.5 pp |
| MDD (%) | -14.02 | -16.23 | -17.60 | -1.37 pp |
| Win Rate (%) | 52.9 | 51.0 | 50.0 | -1.0 pp |

All modification metrics are WORSE than the baseline (highlighted). The modification failed to improve risk management.

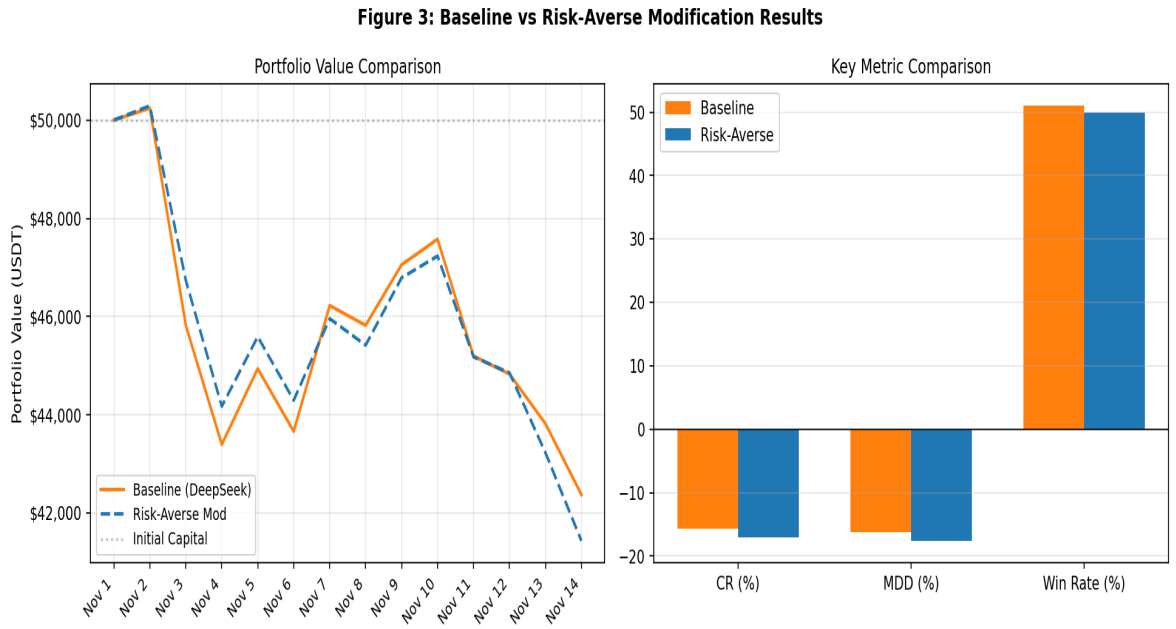


*Figure 3: Left -- Portfolio value trajectories (Baseline vs Risk-Averse vs CD5). Right -- Key metric comparison bars.*

### 5.3 Analysis: Why Did the Modification Perform Worse?

Four mechanisms explain this counterintuitive result:

**Cash reserve reduced deployed capital:** The 10% minimum cash requirement held capital idle. In a bearish market, this reduced exposure to losses; but gains on recovery days were also reduced, leading to a net negative effect.

**35% cap caused concentrated bets:** Instead of spreading across all 10 assets, the agent concentrated in 5 positions. The AVAX-USDT position grew to dominate (~40% of invested capital), dramatically increasing idiosyncratic risk and volatility (59% vs 28% in baseline).

**Hold threshold missed recovery windows:** The 0.5% gain threshold caused the agent to hold through brief price recovery windows that the baseline exploited through active rebalancing.

**Prompt constraints are unreliable for quantitative rules:** The LLM interpreted 'at least 3 assets' loosely (3-5 assets varies per day). Quantitative constraints are better enforced in code logic than in natural language prompts.

**Key Takeaway:** This finding aligns with the paper's observation that 'risk control capability determines cross-market robustness' -- but reveals that prompt-based constraints do not reliably translate into quantitative risk management. Production-grade risk management requires code-level enforcement.

## 6. Debug Diary: Main Blockers + Resolutions

| # | Blocker | Resolution |
|---|---|---|
| 1 | Git checkout failed on Windows (colon in paths) | Bypass checkout: binary re-checkout via 'git show HEAD:path' - bypasses NTFS path validation |
| 2 | UTF-16 BOM file encoding (PowerShell redirection) | Binary re-checkout method produces clean UTF-8 files without BOM |
| 3 | fcntl module not available on Windows | Cross-platform conditional import; no-op lock for Windows (single-threaded, safe) |
| 4 | LangChain MCP tool responses: list vs string content | Monkey-patch format_message_content() to collapse list blocks into joined string |
| 5 | Experiment halted after Day 12 (silent exit) | Resume auto-resumes from last recorded date; ran days 13-14 manually then restarted |
| 6 | Alpha Vantage free tier rate limit | Not critical; agent falls back to price-only analysis when news is unavailable |

## 7. Conclusions

### 7.1 What Successfully Reproduces

- Direction and ranking: DeepSeek-V3.1 outperforms all other models AND the CD5 baseline
- Qualitative findings: All models show negative returns (-12% to -19%) during bearish Nov 1-14 period
- Framework functionality: Full MCP toolchain, position logging, metrics computation work end-to-end
- Win rates: All models cluster at 41-59%, consistent with paper

### 7.2 What Does Not Exactly Reproduce

- Exact numeric values: My run CR=-15.67% vs paper -12.18% (~3.5 pp gap)
- Causes: LLM non-determinism (temperature not fixed), undocumented model checkpoint, different news API responses at different timestamps

### 7.3 Reproducibility Assessment

| Aspect | Level | Notes |
|---|---|---|
| Core qualitative findings | HIGH | Direction, ranking, conclusions all confirmed |
| Framework execution | HIGH | All components run end-to-end on Windows |
| Exact numeric results | LOW | LLM non-determinism; no seeds documented in paper |
| Setup ease | MEDIUM | Windows compatibility fixes required; data pre-packaged |

### 7.4 Recommendations

- Pin model checkpoint versions exactly (full hash, not just name) in paper appendix
- Report temperature and sampling parameters -- critical for LLM agent reproducibility
- Run 3-5 trials and report mean +/- std -- single runs are misleading for stochastic systems
- Test on Windows as well as Linux/macOS -- fcntl and path assumptions affect portability
- The pre-packaged historical price data (crypto_merged.jsonl) is the strongest reproducibility feature -- preserve this practice

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### References

[1] Fan, T., Yang, Y., Jiang, Y., Zhang, Y., Chen, Y., & Huang, C. (2025). AI-Trader: Benchmarking Autonomous Agents in Real-Time Financial Markets. arXiv:2512.10971.

[2] Yao, S., Zhao, J., Yu, D., et al. (2022). ReAct: Synergizing Reasoning and Acting in Language Models. arXiv:2210.03629.

[3] Anthropic. (2025). Model Context Protocol (MCP). https://modelcontextprotocol.io/

[4] DeepSeek-AI. (2025). DeepSeek-V3 Technical Report. arXiv:2412.19437.